CS5295: NETWORK AND CLOUD SECURITY

New Syllabus Proposal

Effective Term
Semester A 2026/27

Part I Course Overview

Course Title
Network and Cloud Security

Subject Code
CS - Computer Science
Course Number
5295

Academic Unit
Computer Science (CS)

College/School
College of Computing (CC)

Course Duration
One Semester

Credit Units
3

Level
P5, P6 - Postgraduate Degree

Medium of Instruction
English

Medium of Assessment
English

Prerequisites
Nil

Precursors
Nil

Equivalent Courses
Nil

Exclusive Courses
Nil

Part II Course Details

Abstract

This course provides students with a comprehensive exploration of security challenges in network and cloud environments. It covers security issues in network protocols (e.g., DNS, BGP, TLS, VPN), advanced topics like intrusion detection systems (IDS), firewalls, and content delivery network (CDN) security, and modern cloud security practices. With a focus on professional and practical skills, students will engage in hands-on labs to better understand the relevant security concepts, leveraging insights from industry leaders like Cloudflare, Palo Alto Networks, and AWS. The course prepares students to design, implement, and manage secure network and cloud architectures for real-world cybersecurity roles.

Course Intended Learning Outcomes (CILOs)

	CILOs	Weighting (if app.)	DEC-A1	DEC-A2	DEC-A3
1	Design secure network architectures to mitigate threats across protocols and infrastructure.	30	x	x	
2	Implement robust security controls for cloud-based systems and applications.	30	x	x	
3	Evaluate advanced detection and reaction solutions for network and cloud threats.	20	x	x	x
4	Develop strategies to secure modern cloud-native technologies and emerging network paradigms.	20	x	x	x

A1: Attitude
Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

A2: Ability
Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

A3: Accomplishments
Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

Learning and Teaching Activities (LTAs)

	LTAs	Brief Description	CILO No.	Hours/week (if applicable)
1	Lecture	Explain core ideas, concepts, principles, and architectures in network and cloud security.	1, 2, 3, 4	2 hours/week
2	Tutorial	Help students understand and apply concepts from lectures through problem-solving and hands-on practical exercises.	1, 2	1 hour/week
3	Assignment	Require students to develop the ability to think in-depth about concepts and apply them to solve complex problems independently.	1, 2, 3, 4	

4	Quiz	Assess students' understanding of key network and cloud security concepts covered in the course.	1, 2, 3, 4	

Assessment Tasks / Activities (ATs)

	ATs	CILO No.	Weighting (%)	Remarks ("-" for nil entry)	Allow Use of GenAI?
1	Assignment 1 Analyze network protocol vulnerabilities and propose mitigation strategies.	1, 3	10	-	Yes
2	Assignment 2 Design and implement cloud security controls for a given scenario.	2, 3	10	-	Yes
3	Assignment 3 Evaluate emerging security technologies and their applications.	3, 4	10	-	Yes
4	Quiz Assess understanding of key network and cloud security concepts.	1, 2, 3, 4	10	-	No

Continuous Assessment (%)

40

Examination (%)

60

Examination Duration (Hours)

2

Minimum Examination Passing Requirement (%)

30

Additional Information for ATs

To pass the course, students must achieve at least 30% of the maximum marks in the examination.

Assessment Rubrics (AR)

Assessment Task

Assignments

Criterion

Ability to apply network and cloud security principles to practical scenarios.

Excellent
(A+, A, A-): High proficiency in designing and implementing security solutions.

Good
(B+, B, B-): Significant understanding with minor errors.

Fair
(C+, C, C-): Moderate competence with some gaps.

Marginal
(D): Basic understanding, limited application.

Failure
(F): Below marginal level.

Assessment Task
Quiz

Criterion
Ability to demonstrate knowledge of network and cloud security concepts.

Excellent
(A+, A, A-): Accurate and comprehensive responses.

Good
(B+, B, B-): Mostly accurate with minor errors.

Fair
(C+, C, C-): Moderate accuracy with notable gaps.

Marginal
(D): Basic knowledge, significant errors.

Failure
(F): Below marginal level.

Assessment Task
Final Exam

Criterion
Ability to explain and apply network and cloud security concepts.

Excellent
(A+, A, A-): Deep understanding and accurate application.

Good
(B+, B, B-): Good grasp with minor inaccuracies.

Fair
(C+, C, C-): Moderate understanding with errors.

Marginal

(D): Basic knowledge, limited depth.

Failure

(F): Below marginal level.

Part III Other Information

Keyword Syllabus

The syllabus evolves with the latest network and cloud security industry trends. Example topics include: network security, protocol vulnerabilities, mitigation strategies, firewall, incident detection, cloud security, shared responsibility, DDoS protection, zero trust, cloud-native security.

Syllabus

a. **Network Security Principles**
   - Securing network protocols and infrastructure
   - Industry-standard mitigation strategies

b. **Intrusion Detection and Prevention**
   - Designing and managing IDS and firewalls
   - Real-time threat response techniques

c. **Cloud Security Foundations**
   - Cloud service models and security responsibilities
   - Implementing access and data protection controls

d. **CDN and Application Security**
   - Securing content delivery and web applications
   - Leveraging industry solutions for performance and protection

e. **Advanced Security Paradigms**
   - Monitoring and securing cloud-native environments
   - Adopting Zero Trust and SASE frameworks

Reading List

Compulsory Readings

	Title
1	Nil

Additional Readings

	Title
1	Stallings, W. (2020). Network Security Essentials: Applications and Standards. Pearson.
2	Cloudflare Documentation: Cloudflare Application Services
3	Palo Alto Networks Resources: Palo Alto Networks Cybersecurity
4	AWS CloudWatch Security: AWS CloudWatch Monitoring Security